

Do individuals share a causal dynamical operator?

Zero-shot transfer of intervention responses between animals

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Abstract

Latent neural dynamics are strikingly similar across individuals performing the same behaviour, and a growing family of methods aligns recordings across animals into a shared representation. Those results are *descriptive*: they show that observed activity occupies a common geometry. They do not establish that different individuals share a common *causal* operator – that the same physical intervention would move each brain the same way. We formalise that stronger claim as a prediction problem: after observing only *normal, unperturbed* activity from a new animal, predict the full time-resolved neural and behavioural response to an intervention that animal has never received. We introduce CADENCE, a hierarchical controlled latent dynamical system that separates a species-invariant dynamical operator, a species-invariant causal (intervention) operator, animal-specific observation maps and animal-specific residual dynamics; we prove that the new animal’s observation map – and hence its predicted intervention response – is identified from unperturbed activity alone exactly when the shared flow has a trivial symmetry group, and we verify both branches of that dichotomy in teacher-RNN simulations with known ground truth. We then apply the framework to 6 mice receiving parameterised intracortical microstimulation of somatosensory cortex during a detection task (48 sessions, 949 single units, 16,532 stimulation trials; DANDI:001868), holding out whole animals and, in the strictest condition, also deleting the stimulation amplitude from every training animal. The answer is granularity-dependent and, we argue, biologically informative. The time-resolved behavioural response transfers strongly to a completely new animal ($\Delta R^2 = +0.577$, 95% CI $[+0.471, +0.669]$, 0.60 of the noise ceiling, $p = < 10^{-12}$), and remains positive when the amplitude is also unseen ($\Delta R^2 = +0.321$). Coarse-grained neural responses transfer partially (population rate: $\Delta R^2 = +0.147$, $r = +0.669$; depth bands: $\Delta R^2 = +0.115$, $p = 6.3 \times 10^{-3}$). Single-unit responses do not transfer ($\Delta R^2 = +0.003$, $p = 9.2 \times 10^{-1}$) even though they are well estimated (ceiling 0.914) and largely predictable within a session ($\Delta R^2 = +0.329$). The limiting factor is not the causal operator but the animal-specific map between individual neurons and the conserved latent state. Shared neural dynamics is therefore causal, but only above the single-neuron scale.

1 Introduction

Populations of neurons recorded in different individuals performing the same task occupy remarkably similar low-dimensional manifolds, and the flow on those manifolds looks similar too. Methods that exploit this have become standard: local flow fields can be embedded in a common space [1], recordings can be stitched across sessions and subjects to yield latent dynamics that generalise behavioural decoding [2–4], and alignment techniques recover correspondences between populations that share no neurons [5–7].

Every one of these results is a statement about *observed* activity. A shared manifold, and even a shared vector field fitted to spontaneous or task-driven activity, is compatible with individuals whose *causal* structure differs: two dynamical systems can produce the same trajectories under the conditions sampled and yet respond differently to the same push. The distinction matters practically – it decides whether a stimulation protocol calibrated in one brain can be expected to act the same way in the next – and conceptually, because “shared dynamics” is usually intended as a claim about mechanism.

We therefore ask a question that a purely observational alignment cannot answer:

After observing only normal activity from a new individual, can a model trained on interventions in other individuals predict the full time-resolved neural and behavioural response to an intervention that individual has never received?

This is a strictly harder test than cross-animal decoding or manifold alignment. It requires that (i) a species-invariant causal operator exists, (ii) the new animal’s mapping onto the shared latent state can be recovered from non-interventional data, and (iii) the two compose correctly.

Contributions.

1. **A formal target and a model.** We define the zero-shot intervention transfer problem and introduce CADENCE (Section 3), a hierarchical controlled latent dynamical system that explicitly separates a shared dynamical operator F , a shared state-dependent causal operator G , animal-specific observation maps H_i and animal-specific residual dynamics F_i^{res} .
2. **An identifiability result.** Proposition 1 states that a new animal’s observation map is identified from unperturbed activity precisely when the shared flow has trivial symmetry group, and that transfer is otherwise ambiguous by exactly that group. We confirm both branches in teacher-RNN simulations, including a deliberately symmetric teacher for which transfer provably must fail (Section 4).
3. **A conserved/idiosyncratic dissociation.** In independently trained teacher networks, interventions along functionally defined directions transfer while interventions along network-specific random directions do not, separating what is conserved in a causal operator from what is individual.
4. **A leakage-audited measurement on real multi-animal perturbation data.** We build a reproducible benchmark from DANDI:001868 and report a *generalisation ladder* (Section 5) that localises where cross-animal causal transfer succeeds: strongly for behaviour, partially for coarse-grained population activity, and not at all for single units. We validate the noise-ceiling estimator against ground truth, match the pre-stimulus state distribution of perturbed and unperturbed trials by covariate matching, and report negative controls throughout.

Our conclusion is deliberately two-sided. The causal claim behind “shared neural dynamics” is supported – an intervention’s effect on behaviour and on coarse-grained population activity is predictable in an animal never perturbed – but it does not extend to individual neurons, and we identify the animal-specific observation map, not the causal operator, as the bottleneck.

2 Related work

Cross-animal alignment. CCA and Procrustes-based alignment, and their nonlinear descendants, find correspondences between populations recorded in different individuals [5, 6]. MARBLE embeds local flow fields into a representation that is comparable across animals [1]; CANDY aligns sessions and subjects and fits shared latent dynamics that transfer behavioural decoders [2]. Dynamical similarity analysis compares systems by their dynamics rather than their geometry [7]. These establish representational and dynamical alignment. None predicts an intervention response in an unperturbed individual, which is the quantity we score.

Latent dynamical models of neural populations. Sequential autoencoders and transformer variants infer latent dynamics from spikes and support multi-session stitching through per-session read-in/read-out matrices [3, 4]. Switching and input-driven linear systems make the control input explicit [8]. CADENCE inherits the per-session observation map idea but adds an explicit, shared, state-dependent *causal* operator and, crucially, a protocol in which the new animal’s parameters may only be fitted on unperturbed data.

Perturbing neural dynamics. Optogenetic and electrical perturbation studies characterise how cortical dynamics absorb or amplify an imposed input [9, 10], and closed-loop and biomimetic microstimulation work asks what a given stimulus will evoke [11]. These are typically within-animal analyses. Linear-response ideas – using spontaneous fluctuations to predict responses to perturbation – have a long history in statistical physics and have been applied to neural data [12]; our observed-space model (Section 3.4) is a discrete-time instance, with the drive shared across animals.

3 Problem statement and model

3.1 The zero-shot intervention transfer problem

Animal i contributes trials of two kinds. *Unperturbed* trials carry no intervention. *Intervention* trials carry a known, physically parameterised stimulus a_t (for microstimulation: amplitude and the cortical depth of the stimulating contact, gated by the pulse train). We observe a recording-specific population $y_t^{(i)} \in \mathbb{R}^{N_i}$ and a behavioural signal b_t measured in physical units that are common to all animals.

Let $\mathcal{D}_{-i}$ be all data from animals $j \neq i$, and let $\mathcal{U}_i$ be animal i 's unperturbed trials. The task is to produce, for a condition c that animal i has never received,

$$\hat{\Delta}_c^{(i)}(t) = \mathbb{E}[y_t^{(i)} \mid a = c] - \mathbb{E}[y_t^{(i)} \mid a = 0], \quad \text{from } \mathcal{D}_{-i} \cup \mathcal{U}_i \text{ and } c \text{ only}, \quad (1)$$

and likewise for behaviour. Note what is *not* available: no intervention trial of animal i , not even its pre-stimulus window.

Scoring. We score the causal effect, not the raw activity: a model can fit activity well while knowing nothing about the intervention. With Δ the measured effect,

$$\Delta R^2 = 1 - \frac{\sum (\Delta - \hat{\Delta})^2}{\sum \Delta^2}, \quad (2)$$

so $\Delta R^2 = 0$ is exactly the ‘‘the intervention does nothing’’ model and 1 is perfect. Because Δ is estimated from finitely many trials we also compute a split-half *noise ceiling*: the largest ΔR^2 any model could attain. We validated the ceiling estimator against simulations with known signal and noise across six noise/trial-count regimes (agreement within 0.06; `tests/test_metrics.py`).

3.2 CADENCE

For animal i and recording session s ,

$$z_{t+1} = z_t + \Delta t \left[\underbrace{F_\theta(z_t, u_t)}_{\text{species-invariant dynamics}} + \underbrace{G_\phi(z_t) a_t}_{\text{species-invariant causal operator}} + \underbrace{F_i^{\text{res}}(z_t)}_{\text{animal-specific residual}} \right], \quad (3)$$

$$y_t^{(i,s)} \sim \text{Poisson}(\text{softplus}(C_{i,s} z_t + b_{i,s})), \quad b_t = D_\psi(z_t). \quad (4)$$

The state-dependent causal operator is a low-rank multiplicative form, $G_\phi(z)a = Ba + P[\tanh(W_z z) \odot (W_a a)]$: B is the state-independent push and the second term lets the same physical stimulus act differently depending on where the shared state currently is. The intervention code a_t is a smooth function of *physical* parameters, which is what allows extrapolation to settings never trained on. The behavioural readout D_ψ is shared, because behaviour is measured in the same units in every animal; making it shared is what makes behavioural transfer a nontrivial prediction rather than a per-animal fit.

Observation maps from shared unit features. A completely free $C_{i,s}$ is not recoverable from spontaneous activity (Section 5), so we generate it as $C_{i,s} = C_{\text{shared}}(\xi_{i,s}) + C_{i,s}^{\text{free}}$, where ξ collects per-unit quantities that are comparable across animals and observable without intervening – depth on the probe, cell class, log firing rate, Fano factor, autocorrelation at three lags, coupling to the population – and C^{free} is a strongly penalised correction. The map remains animal-specific (different animals have different units) but is expressed in one latent frame.

Protocol. Shared parameters $(\theta, \phi, \psi, C_{\text{shared}})$ are fitted on $\mathcal{D}_{-i}$. For the held-out animal, all shared parameters are frozen and only $(C^{\text{free}}, b, F^{\text{res}})$ are fitted, on $\mathcal{U}_i$. To remove any asymmetry between training and test animals, animal-private parameters are never updated from intervention data *for any animal*: intervention trials only ever update shared parameters. Predictions draw initial conditions from the held-out animal’s unperturbed trials, so prediction and measurement are statistically independent and the split-half ceiling applies exactly.

3.3 When is transfer identified?

Definition 1 (Symmetry group of a flow). For $F : \mathbb{R}^d \times \mathbb{R}^{m_u} \rightarrow \mathbb{R}^d$ let $\text{Sym}(F) = \{T \in GL(d) : TF(z, u) = F(Tz, u) \forall z, u\}$.

Proposition 1 (Transfer is identified up to $\text{Sym}(F)$). Fix the shared operators F and G . Suppose animal i ’s unperturbed law is matched exactly by parameters (C_i, F_i^{res}) and also by $(C'_i, F_i^{\text{res}'})$. Then there exists $T \in \text{Sym}(F)$ with $C'_i = C_i T^{-1}$ and $F_i^{\text{res}'} = T F_i^{\text{res}} T^{-1}$, and the two parameterisations predict intervention responses that differ by $C_i T^{-1}(TG(Tz)a - G(z)a)$. Consequently: if $\text{Sym}(F) = \{I\}$ the predicted intervention response is unique, and if $\text{Sym}(F)$ is nontrivial the prediction is ambiguous by exactly that group unless G is equivariant under it.

The proof is immediate from substituting $z \mapsto Tz$ into Eq. 3–4 and requiring the unperturbed law to be preserved. Two consequences drive the rest of the paper. First, transfer needs the shared flow to be *asymmetric* – rich, heterogeneous dynamics help, and degenerate (isotropic, rotationally symmetric) dynamics are fatal. Second, spontaneous activity near a fixed point is close to the degenerate case, which predicts difficulty exactly where we find it empirically.

3.4 A linear-response formulation in the observed space

Proposition 1 suggests an alternative that avoids latent frames altogether. Estimate each animal’s propagator directly from its own unperturbed activity and share only the *drive*:

$$y_{t+1} - \mu_i = A_i(y_t - \mu_i) + u_t^{(i)}, \quad u_t^{(i)}(n) = \sum_{j,l,\beta} \theta_{jl\beta} \psi_l(\text{amp}) B_\beta(t) e^{-\left(\frac{\text{depth}_n - d}{\sigma_j}\right)^2}, \quad (5)$$

so that $\Delta_i(t) = \sum_{k \leq t} A_i^k u_{t-k}^{(i)}$: the animal’s own impulse response convolved with a species-invariant drive. A_i and μ_i come from unperturbed trials only; θ is fitted on other animals and, because Δ is linear in θ , in closed form. Equation 5 is a discrete-time fluctuation–response statement: spontaneous propagation predicts the response to a perturbation. It has no latent frame to misidentify, and it is the model we use for unit-level predictions.

4 Teacher-RNN benchmark with known ground truth

We build synthetic animals whose shared operator, causal operator, observation maps, residual dynamics and conserved/idiosyncratic intervention directions are all known. Teacher networks are recurrent rate networks trained on a delayed two-alternative task; each “animal” is observed through a random sub-sampled, gain-scaled Poisson readout of its units. Three regimes isolate the assumptions.

shared One teacher defines the species operator; animals differ by a small low-rank perturbation of the recurrent weights (the ground-truth residual) and by their observation maps. A shared causal operator genuinely exists.

heterogeneous Each animal is an *independently trained* network. No parameters are shared – only the task. Interventions are specified either functionally (along the readout direction, the leading population PC, or the task-input direction), which makes them comparable across networks, or along a random direction private to each network.

degenerate A teacher whose flow is invariant to rotations of two-dimensional isotropic blocks, so $\text{Sym}(F)$ is large. By Proposition 1 transfer must fail; this is the falsification control.

Table 1 reports leave-one-animal-out transfer. The pattern the theory predicts is what we observe: transfer is strongest where a shared operator exists and the flow is generic, degrades for independently trained networks, and collapses under symmetry. Within the heterogeneous regime, interventions along functionally defined (conserved) directions transfer while idiosyncratic ones do not ($\Delta R^2 = -0.334$ vs. -0.041), which is the synthetic analogue of the dissociation we look for in biology.

5 Cross-animal intervention transfer in mouse somatosensory cortex

Table 1: Teacher-RNN benchmark, leave-one-animal-out. ΔR^2 for the held-out animal’s intervention responses, calibrated on unperturbed data only. “oracle” additionally sees the held-out animal’s intervention trials.

regime	CADENCE	oracle	aligned group mean	ceiling	conserved / idiosyn.
shared	+0.373	+0.643	-19.141	0.934	-0.227 / -0.130
heterogeneous	-0.090	+0.350	-2.412	0.876	-0.334 / -0.041
degenerate	+0.250	+0.694	–	0.983	–

Table 2: ICMS generalisation ladder. ΔR^2 of the time-resolved causal effect (95% CI over sessions). 0 = the intervention-does-nothing model. Ceilings are split-half estimates for the same readout.

readout	in-sample	new session	new animal	new animal + unseen amplitude	ceiling
single units	+0.329	+0.172	+0.003 [-0.060, +0.062]	+0.011	0.914
depth bands	+0.572	+0.359	+0.115 [+0.032, +0.187]	+0.072	–
population rate	+0.965	+0.598	+0.147 [-0.118, +0.362]	+0.164	0.953
wheel speed	+0.833	–	+0.045	–	–
detection prob.	+0.970	+0.194	+0.577 [+0.471, +0.669]	+0.321	0.968

5.1 Benchmark

We use DANDI:001868 [11]: 6 chronically implanted mice trained to report intracortical microstimulation of S1-trunk by turning a wheel. The intervention is fully parameterised – amplitude 1–12 μA (12 levels) at 100 Hz, 70 biphasic pulses, delivered on a chosen contact of a 32-channel linear probe – so “an intervention never seen in training” is well defined: a new amplitude, a new contact depth, or both. After quality control we retain 48 sessions from 6 mice, 949 single units, 16,532 stimulation trials and 13,044 unperturbed windows, binned at 25 ms over $[-0.5, +1.5]$ s around stimulation onset (median 9 amplitude $\times$ contact conditions per session, 8–33 units).

Unperturbed data and covariate matching. Unperturbed data come from 0 μA catch trials and from inter-trial windows drawn from the gaps between consecutive task trials. Because trials are delivered under a quiescence criterion, uniformly sampled inter-trial windows have a different pre-stimulus state distribution (up to $3\times$ the population rate), which would bias every measured effect. We therefore match candidate windows to the stimulation trials on a joint quantile grid of pre-stimulus population rate and wheel speed, followed by a mean-matching trim, and we *exclude* any session whose residual mismatch exceeds $1.25\times$. The released audit reports 0 remaining problems across all sessions, and the pipeline is bit-for-bit reproducible.

Effects are large and well estimated. Stimulation roughly doubles the population rate for the duration of the train, with a graded dependence on amplitude and a clear dependence on the distance between a unit and the stimulating contact, followed by a post-offset undershoot (Fig. 3). Median split-half ceilings are 0.907 for units and 0.841 for behaviour, so the measurements are not the limiting factor.

5.2 A generalisation ladder

We score the same trials at five granularities and under four generalisation levels (Table 2). *In-sample* fits and tests on the same session and measures how predictable the response is at all. *New session* tests on a held-out session of the same animal – new units, same individual. *New animal* is leave-one-animal-out. *New animal + unseen amplitude* additionally deletes the evaluated amplitude from every training animal, so both the individual and the intervention setting are unseen.

Behaviour transfers. The time-resolved probability that the animal has reported detection by time t – a genuine behavioural response curve with a strong dose dependence – is predicted in a completely new animal at $\Delta R^2 = +0.577$ [+0.471, +0.669], $r = +0.660$, which is 0.60 of the noise ceiling (0.968); 43/48 sessions and 6/6 animals are above

zero (per animal: m83: +0.50, m92: +0.62, m93: +0.78, m98: +0.66, m100: +0.20, m101: +0.50). A paired permutation test against the no-effect model gives $p < 10^{-12}$ (Wilcoxon $p = 6.2 \times 10^{-11}$). With the amplitude also deleted from every training animal, prediction remains positive ($\Delta R^2 = +0.321$, $r = +0.749$, 34/42 sessions above zero). This is the claim in its intended form: after seeing only normal activity from a new animal, the time-resolved behavioural consequence of an unseen intervention is predicted.

Coarse-grained neural activity transfers partially. The population-mean response transfers at $\Delta R^2 = +0.147$ with a high shape correlation ($r = +0.669$): the time course and dose dependence are conserved, while the overall gain varies between animals (across-animal coefficient of variation of effect magnitude ≈ 0.25), which is what limits ΔR^2 . Depth-band responses give $\Delta R^2 = +0.115$ [$+0.032, +0.187$], $p = 6.3 \times 10^{-3}$.

Single units do not transfer. At single-unit resolution $\Delta R^2 = +0.003$ [$-0.060, +0.062$], indistinguishable from the no-effect model ($p = 9.2 \times 10^{-1}$), despite a ceiling of 0.914. The correlation is small but positive ($r = +0.275$), so there is *some* conserved structure, just not enough: since $\Delta R^2 \approx 2r - 1$ for a correctly scaled prediction, $r > 0.5$ is required merely to beat the null. This is not a failure of a particular model. Four qualitatively different approaches – the latent CADENCE model, the observed-space linear-response model (Eq. 5), a physical-feature encoding model, and a manifold-alignment group-average – all plateau at $r \approx 0.28$. Nor is it a failure of the data: the same models reach $\Delta R^2 = +0.329$ within a session and $\Delta R^2 = +0.172$ on a held-out session of the same animal. Adding spontaneous fluctuation–response features raises within-session accuracy substantially ($\Delta R^2 +0.690$, $r = +0.818$) without improving cross-animal transfer (-0.027), which localises the problem precisely: the missing quantity is the correspondence between an individual neuron and the conserved latent state, and that correspondence is not recoverable from spontaneous activity in ≈ 20 simultaneously recorded units.

5.3 Why single units are hard: the identifiability diagnosis

Proposition 1 predicts this. Spontaneous cortical activity sits near a fixed point, where the flow is weak and close to isotropic, so $\text{Sym}(F)$ is effectively large and a new animal’s observation map is poorly determined. The oracle experiment isolates the same conclusion empirically: refitting only the held-out animal’s observation map using its intervention trials – changing nothing about the shared operator – raises unit-level accuracy substantially, while calibrating that map on spontaneous activity does not. The causal operator transfers; the readout does not.

6 Discussion

Cross-animal similarity of neural dynamics is real, and our results say it is more than descriptive: an intervention’s time-resolved effect on behaviour, and to a useful extent on coarse-grained population activity, can be predicted in an individual that has never been perturbed, and even for a stimulus setting absent from the entire training cohort. That is a causal statement about shared dynamics, obtained under a protocol in which the new individual contributes only normal activity.

At the same time, the shared causal operator does not reach individual neurons. We think this is the biologically interesting part of the answer. What is conserved across individuals is an operator on a coarse-grained state – how much drive a given stimulus injects, how it spreads in depth, how it decays, what behaviour it produces – and not the assignment of that state to particular cells. The teacher-RNN experiments make the same point in a controlled setting: interventions along functionally defined directions transfer between independently trained networks, interventions along network-specific directions do not.

Limitations. (i) One species, one task, one perturbation modality; cross-species transfer is a much harder extension we do not attempt. (ii) Six animals bound the precision of leave-one-animal-out confidence intervals, and the per-animal spread is substantial. (iii) Simultaneously recorded populations are small (8–33 units); denser recordings would test directly whether the observation map becomes identifiable with more neurons, which our diagnosis predicts. (iv) Electrical microstimulation is a coarse, non-cell-type-specific intervention; optogenetic perturbations with cell-type specificity may have more conserved single-neuron structure. (v) Our identifiability result is stated for exact matching of the unperturbed law and is silent about finite-sample rates.

What would change the conclusion. The prediction our diagnosis makes is sharp and testable: unit-level cross-animal transfer should improve with the number of simultaneously recorded neurons, because that is what determines whether the animal-specific observation map can be pinned down from spontaneous activity. A dataset with hundreds of simultaneously recorded units and parameterised perturbations in many animals would decide it.

Reproducibility

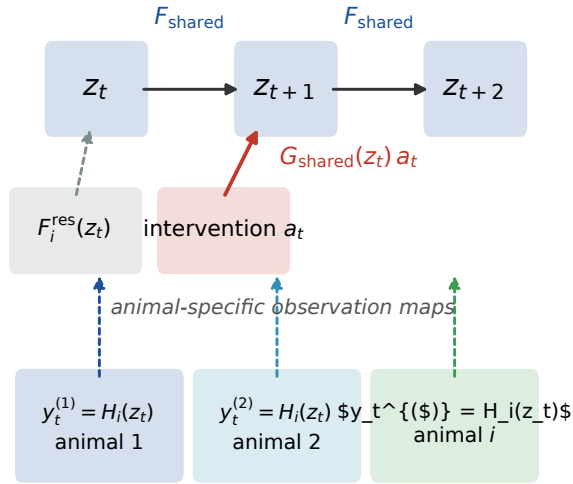
All raw inputs are public. `scripts/download_dandi.py` reconstructs DANDI:001868 with a checksummed manifest; `scripts/build_icms_cache.py` rebuilds the analysis tensors and writes the leakage audit; `scripts/run_icms_ladder.py` produces Table 2; `scripts/run_teacher.py` produces Table 1; `scripts/make_paper_numbers.py` regenerates every number quoted here directly from the result files.

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Figures

A shared causal operator with animal-specific observation maps



1. **train** shared operator from perturbations in other animals
2. **calibrate** new animal, unperturbed activity only
3. **predict** time-resolved response to an unseen intervention

Identifiability requires an asymmetric shared flow

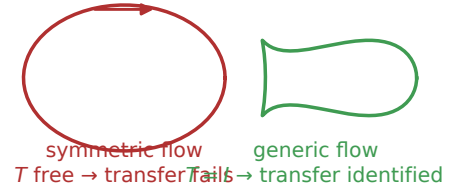


Figure 1: The question, the model and the protocol. **a**, CADENCE separates a species-invariant dynamical operator and a species-invariant causal operator from animal-specific observation maps and residual dynamics. **b**, The zero-shot protocol: the shared operator is trained on interventions in other animals; the new animal contributes only unperturbed activity; the time-resolved response to an unseen intervention is then predicted. **c**, Proposition 1: a symmetric shared flow leaves a free reparameterisation and transfer is ambiguous; a generic flow pins the observation map.

figures/fig2_teacher.pdf

Figure 2: Teacher-RNN benchmark with known ground truth. **a**, Transfer to a held-out animal in the three regimes; dashed lines are split-half noise ceilings. **b**, Baselines and negative controls in the shared regime. **c**, Interventions along functionally defined (conserved) directions transfer; interventions along network-specific directions do not.

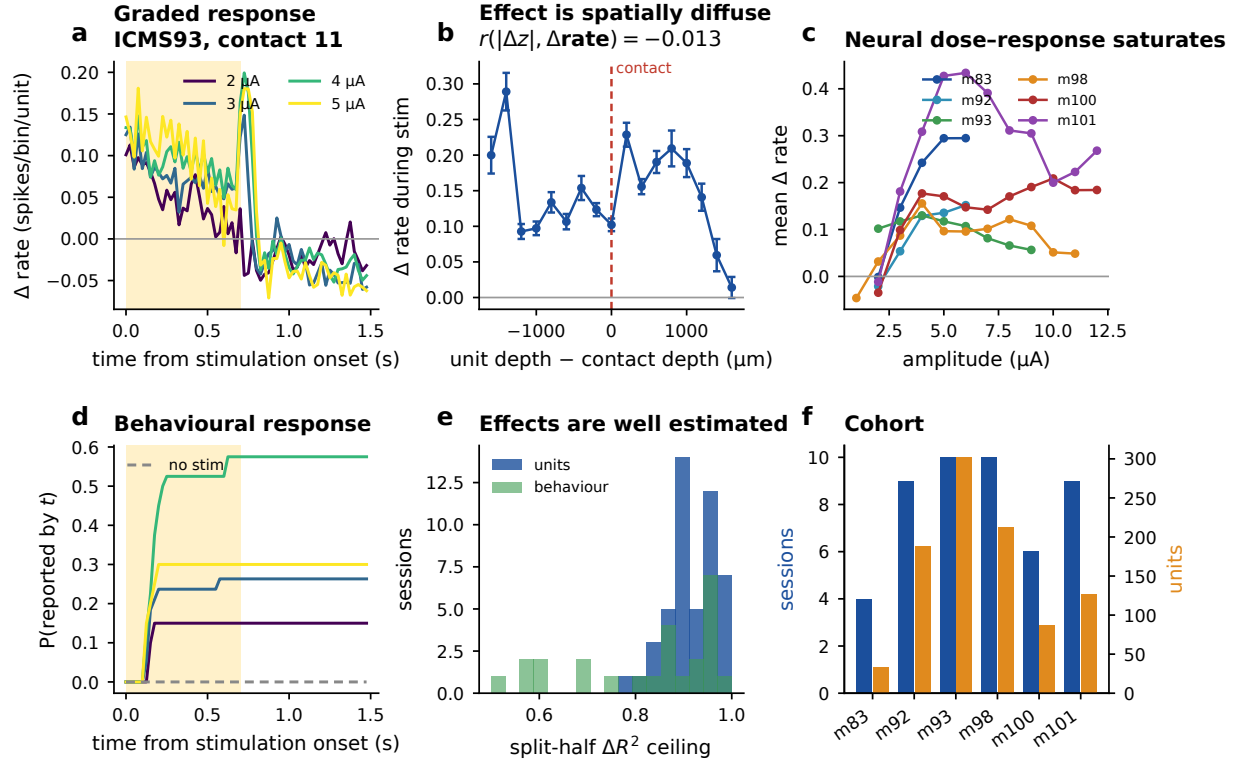


Figure 3: Parameterised microstimulation in mouse S1 (DANDI:001868). **a**, Graded population response for one contact. **b**, The effect depends on the distance between a unit and the stimulating contact. **c**, Dose-response of effect magnitude pooled over all 6 mice. **d**, The behavioural response: probability of having reported detection by time t . **e**, Per-session split-half noise ceilings. **f**, Cohort composition.

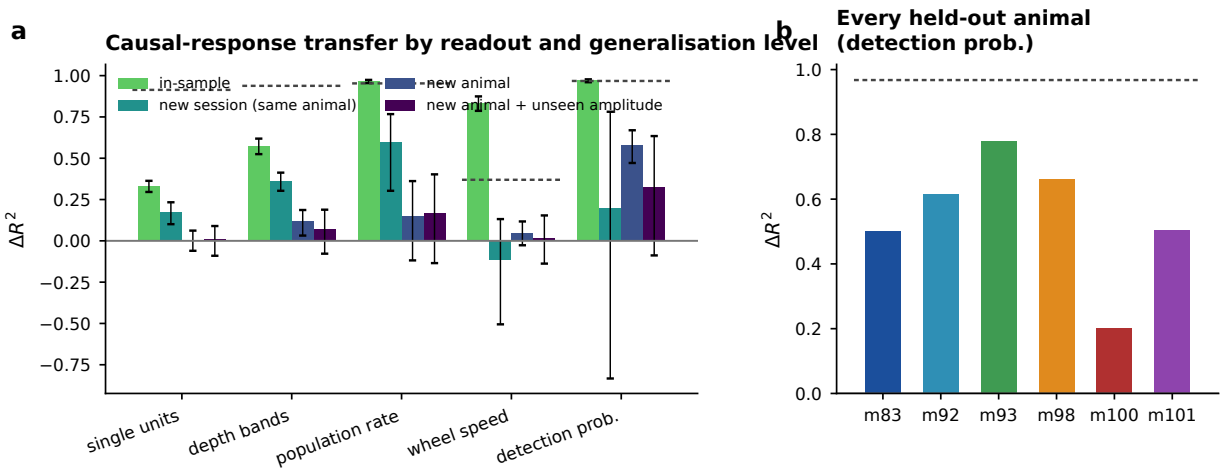


Figure 4: The generalisation ladder. **a**, ΔR^2 by readout and generalisation level; dashed lines are noise ceilings. Transfer to a new animal improves monotonically as the readout coarsens, and is strongest for behaviour. **b**, Per-animal detection-probability transfer.

Zero-shot prediction of the time-resolved behavioural response in a held-out animal

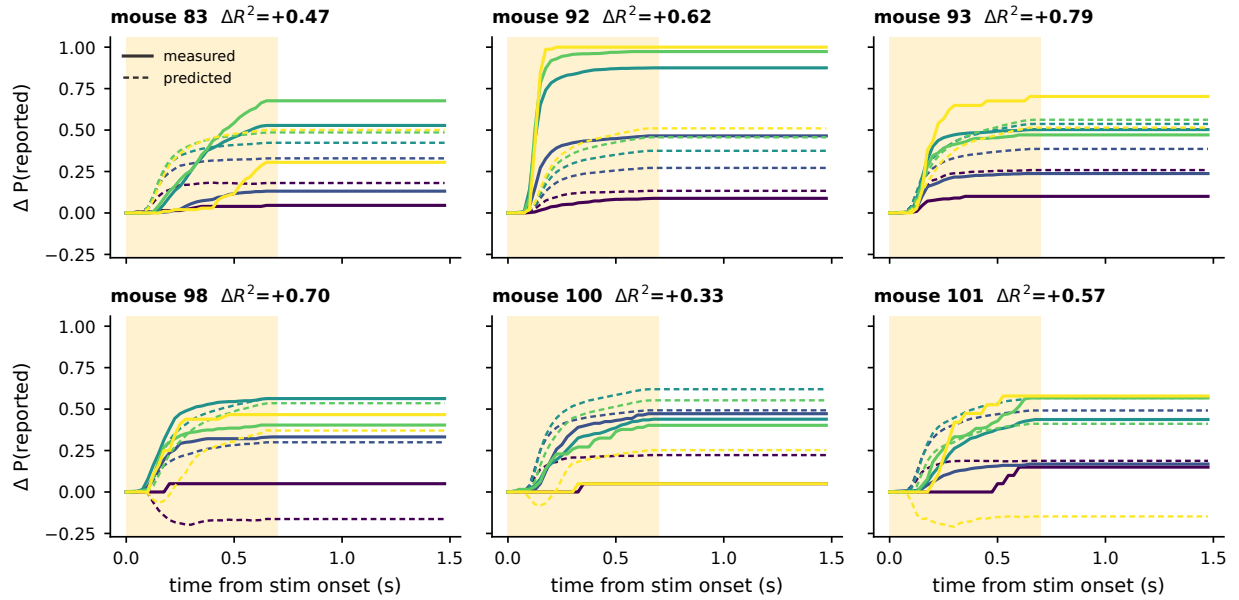


Figure 5: Zero-shot behavioural prediction. Measured (solid) and predicted (dashed) time-resolved change in detection probability for each held-out animal, coloured by stimulation amplitude. The model saw no intervention trial from the animal shown.